

BIOLOGY (THEORY) – MOCK TEST 1

Expert Curated Repeat Questions (Sets 57/1/1, 57/5/1, 57/7/1)

Time: 3 Hours

Max Marks: 70

By Pragya Thunderstudy

General Instructions: (i) 33 questions total. (ii) Sect A: MCQ (1-16). (iii) Sect B: VSA (17-21). (iv) Sect C: SA (22-28). (v) Sect D: Case-based (29-30). (vi) Sect E: LA (31-33).

SECTION A (16 Marks)

1. Hormone secreted by human placenta for pregnancy maintenance?

(A) Relaxin (B) hCG (C) Oxytocin (D) Prolactin

[Source: 57/1/1, 57/5/1]

2. Hind II recognizes a specific sequence of:

(A) Six base pairs (B) Four base pairs (C) Five base pairs (D) Three base pairs

[Source: 57/1/1, 57/5/1/22]

3. Histone core in a nucleosome is an:

(A) Octamer (B) Hexamer (C) Tetramer (D) Pentamer

[Source: 57/5/1]

4. Organism used as biofertilizer for soybean crop:

(A) Azospirillum (B) Rhizobium (C) Nostoc (D) Azotobacter

[Source: 57/1/1, 57/7/1]

5. Rate of formation of new organic matter by consumers:

(A) Secondary productivity (B) NPP (C) GPP (D) Net productivity

[Source: 57/1/1]

6. Not an example of aneuploidy:

(A) Turner's (B) Down's (C) Phenylketonuria (D) Klinefelter's

7. Colostrum is rich in which antibody?

(A) IgE (B) IgA (C) IgG (D) IgM

8. Correct regarding PCR:

(A) Thermostable DNA polymerase (B) No primers (C) DNA separation (D) Inside cell

9. In transcription unit, promoter is towards:

(A) 5' structural (B) 3' structural (C) 5' template (D) 3' coding

10. Immunosuppressive agent produced by:

(A) *Trichoderma polysporum* (B) *Monascus purpureus* (C) *S. cerevisiae* (D) *Penicillium*

11. Palindromic sequence:

(A) 5'-GAATTC-3' (B) 5'-GGGACC-3' (C) 5'-CCTAGG-3' (D) 5'-ATCGCTA-3'

12. Detect HIV infection:

(A) PCR (B) ELISA (C) MRI (D) Widal

13-16: A/R Type. A: Birds males determine sex. R: Males (ZZ) Females (ZW).

14: A: Amniocentesis misused. R: Sex determination female foeticide.

15: A: *Agrobacterium* dicot pathogen. R: Delivers T-DNA.

16: A: DNA fingerprinting relies on Repetitive seq. R: Rep seq are highly conserved.

SECTION B (10 Marks)

17. Interaction between **Fig tree and Wasp**. Why is it Mutualism?

18. Convention for naming **EcoRI**. Write its palindromic sequence.

19. Name two Gonadotropins and their role in **Spermatogenesis**.

20. (a) What are "**Flocs**"? (b) Differentiate **Primary** vs **Activated** Sludge.

21. Two female VDs and two complications if left untreated.

SECTION C (21 Marks)

22. Mode of action of **Oral Pills** and **IUDs (Copper-T)**.

23. Describe **S.L. Miller experiment**. Hypothesis scientists and conclusion.

24. DNA: 5'-ATG AGG GGG ATG-3'. (a) mRNA sequence (b) Amino acids (c) Effect of 'A' deletion.

25. **Pyramid of Biomass** (Sea): (a) Upright or inverted? Why? (b) Name producers/consumers.

26. (a) What are **Transgenic Animals**? (b) Role in Vaccine Safety & Bios (Rosie).

27. Processing **hnRNA** to **mRNA** (Splicing, Capping, Tailing).

28. Syndrome: Short stature, furrowed tongue, open mouth. (a) Identify (b) Cause (c) Autosomal/Sex-linked?

SECTION D (8 Marks)

29. Case Study: Menstrual Cycle.

(a) Identify Ovulatory phase & hormone surge. (b) Endometrium in Luteal phase. (c) Why menstruation occurs?

30. Case Study: Cloning Vectors (pBR322).

(a) Selectable markers function. (b) BamHI insert resistance inactivation. (c) Blue-white screening principle.

SECTION E (15 Marks)

31. (a) Male Gametophyte development with diagrams. (b) Anatropous ovule structure.

OR

(a) Sectional view of Human Ovary follicles. (b) Describe Oogenesis events.

32. (a) Hershey and Chase experiment: Infection, Blending, Centrifugation & Conclusion.

OR

(b) Meselson and Stahl experiment for semi-conservative replication.

33. (a) STP biological treatment: BOD, Floccs, Activated Sludge.

OR

(b) Bt Cotton formation: Cry gene, insect specificity, and activation mechanism.